

Linux Server Security: Initial Hardening for Ubuntu 24.04

Author: Hunain • Release: Ubuntu 24.04 LTS • Reading Time: 6 Min Practical Guide

⚠ The Reality of a Fresh Server: Under Attack in Under 10 Minutes

The moment you launch a new VPS or cloud server, automated internet bots start knocking. Within minutes, scripts begin firing thousands of random passwords at your `root` login. If you leave default passwords and open ports active, it is not a matter of *if* your server gets compromised—it is a matter of *when*.

The Solution: In this step-by-step handbook, we will lock down your server in simple, practical steps: creating an accountable sudo user, enabling unbreakable Ed25519 SSH keys, shutting out root access, setting up a firewall, and deploying Fail2ban.

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1 Create a Dedicated Sudo User

Logging in directly as `root` is dangerous. Every brute-force bot online specifically targets `root`. By creating a custom user with administrative privileges, we add an extra layer of privacy, and every command executed is logged with an exact username.

CODE

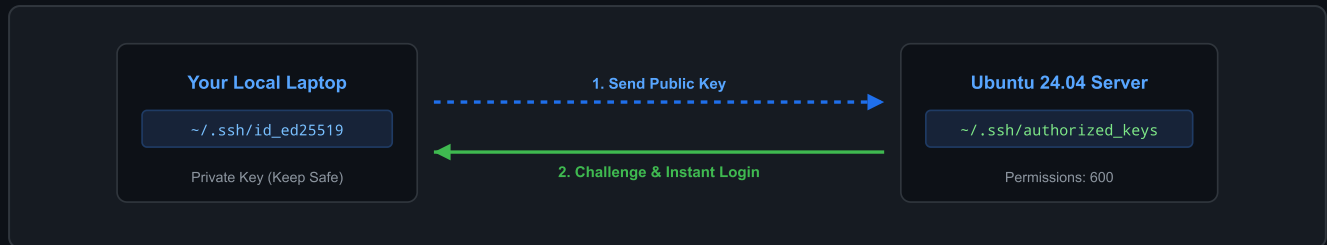
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```
# Log in as root, create your user, and add them to the sudo group
# adduser Hunain
# usermod -aG sudo Hunain
```

Why this matters: Once added to the `sudo` group, user **Hunain** can perform administrative tasks safely by prefixing commands with `sudo`. Every action is recorded in `/var/log/auth.log`, creating an audit trail.

2 Set Up Ed25519 SSH Key Authentication

Passwords can be leaked, guessed, or brute-forced. SSH keys solve this problem by using cryptographic key pairs. **Ed25519** is currently the fastest and most secure elliptic-curve algorithm available—vastly superior to older RSA keys.



CODE (LOCAL MACHINE & SERVER)

Copy

```
# 1. On your personal computer: Generate key with 100 extra encryption rounds
locals$ ssh-keygen -t ed25519 -a 100 -f ~/.ssh/id_ed25519_myserver

# 2. On the server (as Hunain): Store the public key with secure permissions
Hunain@server$ mkdir -p ~/.ssh && chmod 700 ~/.ssh
Hunain@server$ nano ~/.ssh/authorized_keys # Paste your public key here
Hunain@server$ chmod 600 ~/.ssh/authorized_keys

# 3. Test key login from your local computer BEFORE disabling passwords:
locals$ ssh -i ~/.ssh/id_ed25519_myserver Hunain@<SERVER_IP>
```

3 Harden the OpenSSH Daemon

Now that key authentication is working, we lock down SSH by editing `/etc/ssh/sshd_config`. This completely turns off password prompts and blocks root login.

CONFIG (/etc/ssh/sshd_config)

Copy

```
# Turn off root login
PermitRootLogin no

# Disable password logins (keys only!)
PasswordAuthentication no

# Allow ONLY Hunain to connect
AllowUsers Hunain
AllowGroups sudo

# Enforce modern secure key exchange algorithms
KexAlgorithms curve25519-sha256,diffie-hellman-group-exchange-sha256
```

Important Warning: Keep your current terminal open! Apply changes with `sudo systemctl restart ssh` and test in a second terminal. If something goes wrong, you still have your session open to fix it.

4 Configure UFW Firewall

Close all unused network doors. By default, block every incoming connection except SSH and your web ports:

CODECopy

```
# Default policies
# sudo ufw default deny incoming
# sudo ufw default allow outgoing

# Allow essential services
# sudo ufw allow ssh
# sudo ufw allow 80/tcp
# sudo ufw allow 443/tcp

# Enable and check
# sudo ufw enable
# sudo ufw status verbose
```

5 Install & Set Up Fail2ban

Fail2ban scans your system logs. If an IP fails to log in 3 times within 10 minutes, it is banned automatically for 1 hour:

CODECopy

```
# Install fail2ban
# sudo apt update && sudo apt install -y fail2ban

# Create /etc/fail2ban/jail.local
[sshd]
enabled = true
port = ssh
maxretry = 3
findtime = 600
bantime = 3600

# Restart & inspect
# sudo systemctl restart fail2ban
# sudo fail2ban-client status sshd
```

✓ What These Steps Protect Against

Threat / Attack Type	Applied Defense	Real-World Result
Brute-Force SSH Attacks	Ed25519 Keys + Fail2ban	Password dictionary tools are completely blocked.
Direct Root Compromise	PermitRootLogin no	Default automated root scans fail instantly.
Port Probing & Sneak Attacks	UFW Firewall Rules	All ports are closed except explicitly allowed ports.
Stolen Passwords / Leaks	Password Auth Disabled	Passwords cannot be used even if leaked.

🚀 Next Steps: Ready for Production

Your foundational security is complete! From here, enable automatic security updates (`sudo apt install unattended-upgrades`), set up a VPN such as WireGuard if internal administration is required, and configure periodic automated backups.